

◆ SESSION 8 — STUDENT NOTES

Security in Networking

Security Groups • NACLs • Open Port Risks • Misconfiguration Scenarios

◆ SESSION OVERVIEW

Cloud networking security is one of the most critical areas in modern cloud computing.

Organizations must protect their infrastructure from:

- Unauthorized access
- Internet-based attacks
- Misconfigurations
- Data breaches
- Malware and ransomware
- Internal and external threats

In AWS, two major networking security controls are:

- ✓ Security Groups
- ✓ Network ACLs (NACLs)

Understanding how they work is essential for building secure cloud environments.

◆ LEARNING OBJECTIVES

By the end of this session, students should be able to:

- ✓ Explain Security Groups
 - ✓ Explain Network ACLs
 - ✓ Compare SGs and NACLs
 - ✓ Understand open port risks
 - ✓ Identify dangerous misconfigurations
 - ✓ Apply enterprise security best practices
 - ✓ Understand layered network defense
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◆ SECTION 1 — SECURITY GROUPS

WHAT IS A SECURITY GROUP?

A Security Group is a virtual firewall attached directly to AWS resources such as:

- EC2 instances
- Databases
- Load balancers

Security Groups control:

- ✓ Inbound traffic
 - ✓ Outbound traffic
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◆ CHARACTERISTICS OF SECURITY GROUPS

KEY FEATURES

- ✓ Operate at instance level
 - ✓ Stateful firewall
 - ✓ Allow rules only
 - ✓ Automatically allows return traffic
 - ✓ Multiple rules can exist
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◆ WHAT DOES STATEFUL MEAN?

A stateful firewall remembers connections.

Example:

If inbound SSH traffic is allowed:

Port 22 → Allowed Inbound

Return traffic is automatically allowed.

No separate outbound rule is required.

◆ COMMON SECURITY GROUP PORTS

Port	Protocol	Purpose
22	SSH	Linux Administration
80	HTTP	Web Traffic
443	HTTPS	Secure Web Traffic
3389	RDP	Windows Administration
3306	MySQL	Database Access
5432	PostgreSQL	Database Access

◆ EXAMPLE — INSECURE SECURITY GROUP

```
Inbound Rules:  
- Port: 22  
  Source: 0.0.0.0/0
```

◆ WHY IS THIS DANGEROUS?

PROBLEM

0.0.0.0/0 = Entire Internet

This means anyone on the internet can attempt to connect.

RISKS

- Brute-force attacks
 - Credential theft
 - Server compromise
 - Malware deployment
 - Unauthorized access
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◆ SECURE SECURITY GROUP EXAMPLE

Inbound Rules:
- Port: 22
Source: 197.210.10.15/32

◆ WHY THIS IS BETTER

- ✓ SSH restricted to trusted IP
 - ✓ Reduced attack surface
 - ✓ Better governance
 - ✓ Lower security risk
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◆ SECTION 2 — NETWORK ACLs (NACLs)

WHAT IS A NACL?

A Network ACL (Access Control List) is a firewall that operates at subnet level.

NACLs control traffic entering and leaving an entire subnet.

◆ CHARACTERISTICS OF NACLs

- ✓ Operate at subnet level
 - ✓ Stateless firewall
 - ✓ Allow and Deny rules
 - ✓ Rules processed numerically
 - ✓ Return traffic must be explicitly allowed
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◆ WHAT DOES STATELESS MEAN?

A stateless firewall does NOT remember connections.

If inbound traffic is allowed:

- ✓ Outbound traffic must ALSO be allowed manually.

◆ NACL EXAMPLE

Rule 100:
Protocol: TCP
Port: 443
Action: ALLOW
CIDR: 10.0.0.0/16

◆ SECURITY GROUPS VS NACLs

Feature	Security Groups	NACLs
Level	Instance	Subnet
Stateful	Yes	No
Allow Rules	Yes	Yes
Deny Rules	No	Yes
Return Traffic	Automatic	Manual
Scope	Resource Protection	Subnet Protection

◆ ENTERPRISE SECURITY MODEL

Organizations use BOTH Security Groups and NACLs together.

WHY?

This creates:

- ✓ Layered defense
 - ✓ Better segmentation
 - ✓ Stronger governance
 - ✓ Reduced exposure risk
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◆ SECTION 3 — OPEN PORT RISKS

WHAT IS AN OPEN PORT?

An open port allows network traffic into a system.

Some ports are necessary.

Others become dangerous if exposed publicly.

◆ HIGH-RISK PORTS

Port	Risk
22	SSH attacks
3389	RDP attacks
3306	Database exposure
5432	Database exposure

◆ COMMON ATTACK TYPES

BRUTE-FORCE ATTACK

Attackers repeatedly guess passwords.

PORT SCANNING

Attackers scan the internet searching for open services.

REMOTE EXPLOITATION

Attackers exploit vulnerable services.

DATA EXFILTRATION

Sensitive data is stolen from systems.

◆ CRITICAL SECURITY RULE

Never expose administrative ports publicly unless absolutely necessary.

Examples:

- ✗ Port 22 → Public SSH
- ✗ Port 3389 → Public RDP

◆ SECTION 4 — MISCONFIGURATION SCENARIOS

Scenario 1 — Public SSH Exposure

CONFIGURATION

```
Port: 22  
Source: 0.0.0.0/0
```

◆ WHAT HAPPENS?

1. Attackers scan the internet
2. Open SSH is discovered
3. Brute-force attacks begin
4. Credentials may be compromised
5. Infrastructure becomes vulnerable

◆ ENTERPRISE FIXES

- ✓ Restrict SSH access
- ✓ Use VPNs
- ✓ Use Bastion Hosts
- ✓ Enable MFA
- ✓ Monitor logs continuously

◆ Scenario 2 — Public Database Exposure

CONFIGURATION

Database:
PubliclyAccessible: true

◆ RISKS

- Database discovery
 - Data theft
 - Compliance violations
 - Financial penalties
 - Ransomware attacks
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◆ ENTERPRISE FIXES

- ✓ Use private subnets
 - ✓ Restrict inbound traffic
 - ✓ Enable encryption
 - ✓ Use monitoring tools
 - ✓ Apply least privilege
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◆ SECTION 5 — AWS CONSOLE SECURITY REVIEW

HOW CLOUD ENGINEERS REVIEW NETWORK SECURITY

In real AWS environments, cloud engineers review:

- ✓ Security Group rules
 - ✓ Open ports
 - ✓ Allowed IP addresses
 - ✓ Public vs private resources
 - ✓ Network access permissions
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◆ WHAT STUDENTS SHOULD LOOK FOR

SAFE CONFIGURATIONS

- ✓ Restricted administrative access
 - ✓ HTTPS enabled
 - ✓ Private databases
 - ✓ Limited inbound traffic
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DANGEROUS CONFIGURATIONS

- ✗ Port 22 open to everyone
 - ✗ Public databases
 - ✗ Unrestricted RDP access
 - ✗ Entire internet allowed
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◆ EXAMPLE — SAFE WEB SERVER ACCESS

HTTPS (Port 443) → Allowed
SSH (Port 22) → Restricted

◆ WHY HTTPS IS IMPORTANT

Port 443 supports secure encrypted web traffic.

HTTPS:

- ✓ Encrypts communication
 - ✓ Protects sensitive data
 - ✓ Secures websites and applications
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◆ SECTION 6 — ENTERPRISE BEST PRACTICES

NETWORK SECURITY BEST PRACTICES

- ✓ Apply least privilege
 - ✓ Restrict administrative ports
 - ✓ Separate public and private resources
 - ✓ Use layered defense
 - ✓ Monitor traffic continuously
 - ✓ Review firewall rules regularly
 - ✓ Conduct governance audits
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◆ GOVERNANCE PRINCIPLE

Security is not a one-time task.

It is:

- Continuous
 - Monitored
 - Audited
 - Governed
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◆ LAB REVIEW QUESTIONS

1. What is a Security Group?
 2. What is a NACL?
 3. What is the difference between stateful and stateless?
 4. Why is Port 22 risky?
 5. What does 0.0.0.0/0 mean?
 6. Why should databases remain private?
 7. Why is layered defense important?
 8. Why is continuous monitoring important?
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◆ QUICK KNOWLEDGE CHECK

QUESTION 1

Which service operates at subnet level?

- A. Security Group
- B. NACL

✓ Correct Answer: B

QUESTION 2

Which firewall type is stateful?

- A. Security Group
- B. NACL

✓ Correct Answer: A

QUESTION 3

Which configuration is dangerous?

Port: 22
Source: 0.0.0.0/0

✓ Correct Answer: Dangerous

◆ KEY TAKEAWAYS

- ✓ Security Groups protect instances
 - ✓ NACLs protect subnets
 - ✓ Open ports increase risk
 - ✓ Misconfigurations cause breaches
 - ✓ Layered security improves resilience
 - ✓ Governance strengthens cloud security
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◆ FINAL MESSAGE

Great cloud architects do not only deploy infrastructure.

They:

- Analyze risk
- Apply governance

- Standardize security
- Protect enterprise environments

Secure networking is the foundation of resilient cloud architecture.